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1. Introduction

Attendance systems in educational institutions often suffer from inefficiencies and security weaknesses. Manual methods are time-consuming and prone to proxy attendance, while many digital systems lack protection against QR sharing, replay attacks, and token manipulation.

This project proposes a Secure QR-Based Attendance Management System developed using Secure Software Development Lifecycle (SSDLC) principles. The system ensures secure attendance marking through time-limited QR codes, nonce-based validation, and strict backend verification. The goal is to provide a reliable, tamper-resistant, and secure attendance solution.

2. Problem Statement

Current attendance systems face the following issues:

- Proxy attendance
- Screenshot reuse of QR codes
- Replay attacks
- Weak enrollment validation
- Lack of session expiration
- Poor backend security controls

These vulnerabilities reduce trust in attendance records. A secure system is required to ensure authenticity, integrity, and controlled access.

3. Proposed System

The system consists of three main components: Teacher Interface, Student Interface, and Backend Server.

Session Creation (Teacher Side)

The teacher logs in and starts an attendance session. The system generates a QR code containing:

- Session ID
- Course ID
- Timestamp

- Random nonce

The QR code is valid only for a limited time window.

QR Scanning (Student Side)

Students log in using university credentials and scan the QR code. The system sends:

- Student ID
- QR token data
- Device timestamp

All validation is performed on the backend.

Backend Verification

The server checks:

- Student enrollment in the course
- Token validity and expiration
- Nonce reuse
- Duplicate attendance marking
- Time window compliance

If all checks pass, the student is marked Present. If not scanned within the allowed time, the student is marked Absent. The teacher can close the session, after which the token becomes invalid and records are locked.

4. Security Features

The system integrates the following security mechanisms:

- Time-bound QR tokens to prevent replay attacks
- Random nonce to prevent token reuse
- Server-side validation to prevent client manipulation
- Enrollment verification for authorized access
- Role-based access control (Teacher/Student)
- Secure HTTPS communication
- Attendance record locking after session closure

These measures ensure confidentiality, integrity, and availability of attendance data.

5. Technology Stack

The following technologies will be used:

- Frontend: HTML/CSS with Bootstrap for responsive UI
- Backend: Flask
- Database: PostgreSQL
- Authentication: JWT-based authentication
- QR Code Generation: Secure token-based QR encoding library
- Deployment: Cloud-hosted server (e.g., AWS or local VM for prototype)

This stack ensures scalability, security, and maintainability.

6. Secure Software Development Approach

The project follows SSDLC phases:

1. Requirement Analysis
2. Threat Modeling
3. Secure System Design
4. Secure Coding Practices
5. Security Testing (replay tests, duplicate attempts, expired token testing)
6. Secure Deployment

Security is integrated at every phase rather than added after development.

7. Conclusion

The Secure QR-Based Attendance Management System provides a secure and efficient alternative to traditional attendance methods. By using time-bound tokens, nonce validation, backend verification, and SSDLC principles, the system minimizes fraud, prevents replay attacks, and ensures reliable attendance recording.

The project demonstrates practical implementation of secure software development concepts in a real-world academic scenario.